

Unit - 1

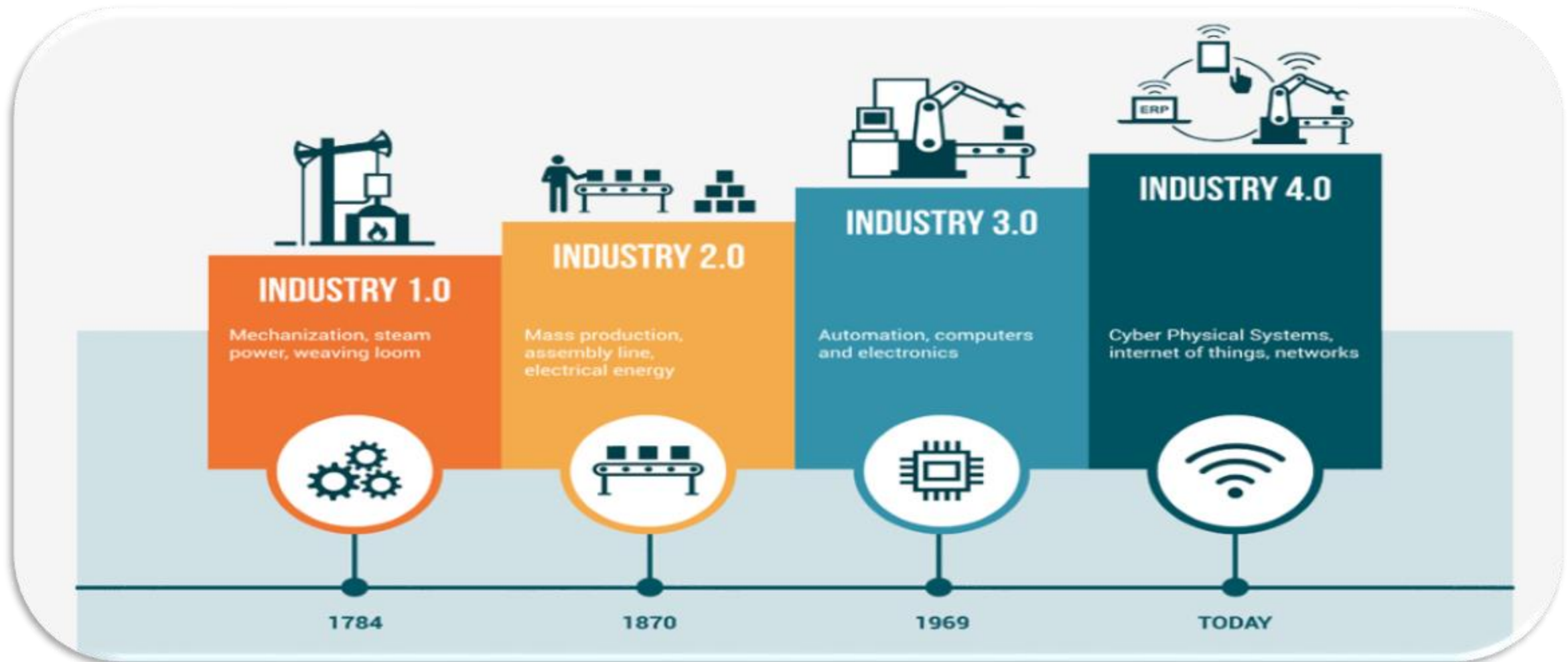
Internet of Things

Market, Opportunities and use cases

Objective

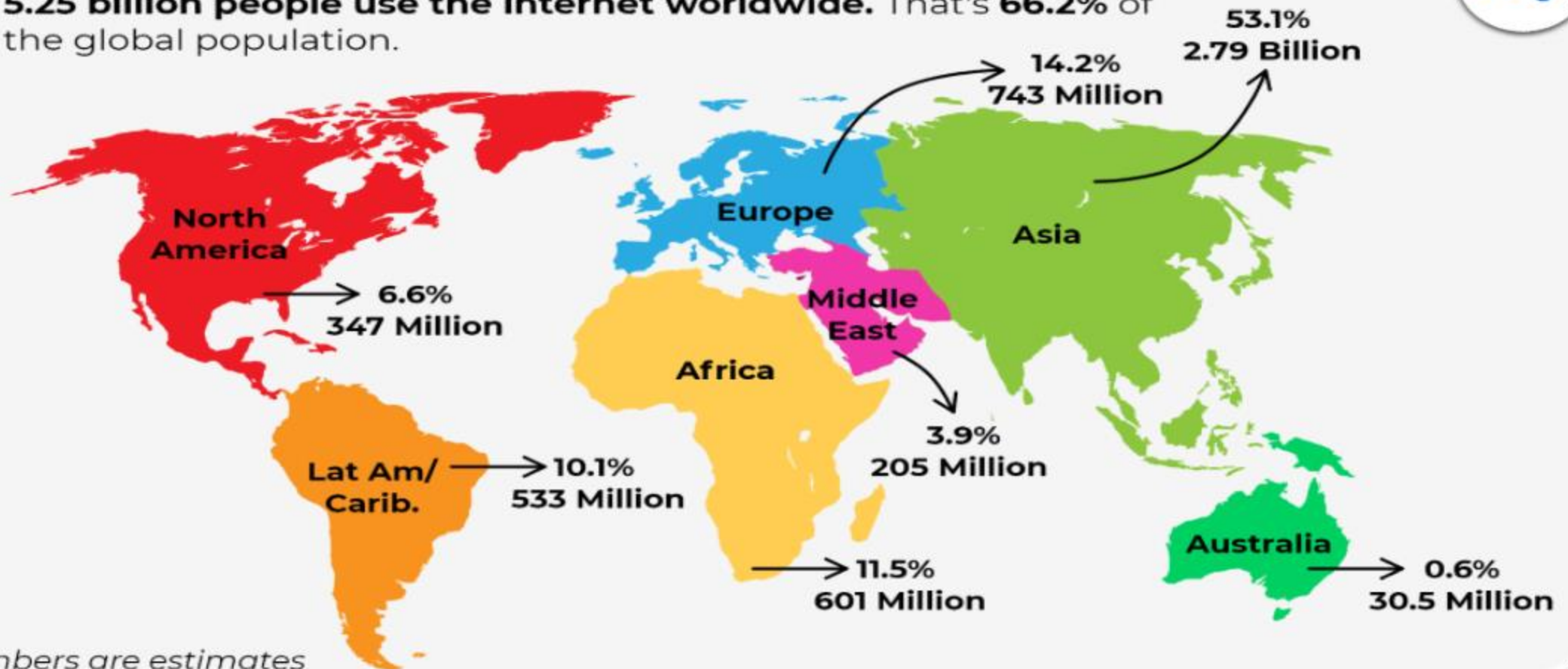
- Industrial Revolution
- Internet Usage and Population Statistics
- IoT Architecture
- What is Internet of Things?
- Why IoT?
- IOT in Technical Perspective
- Where IOT is used? How?
- Future of IoT

Industrial Revolution

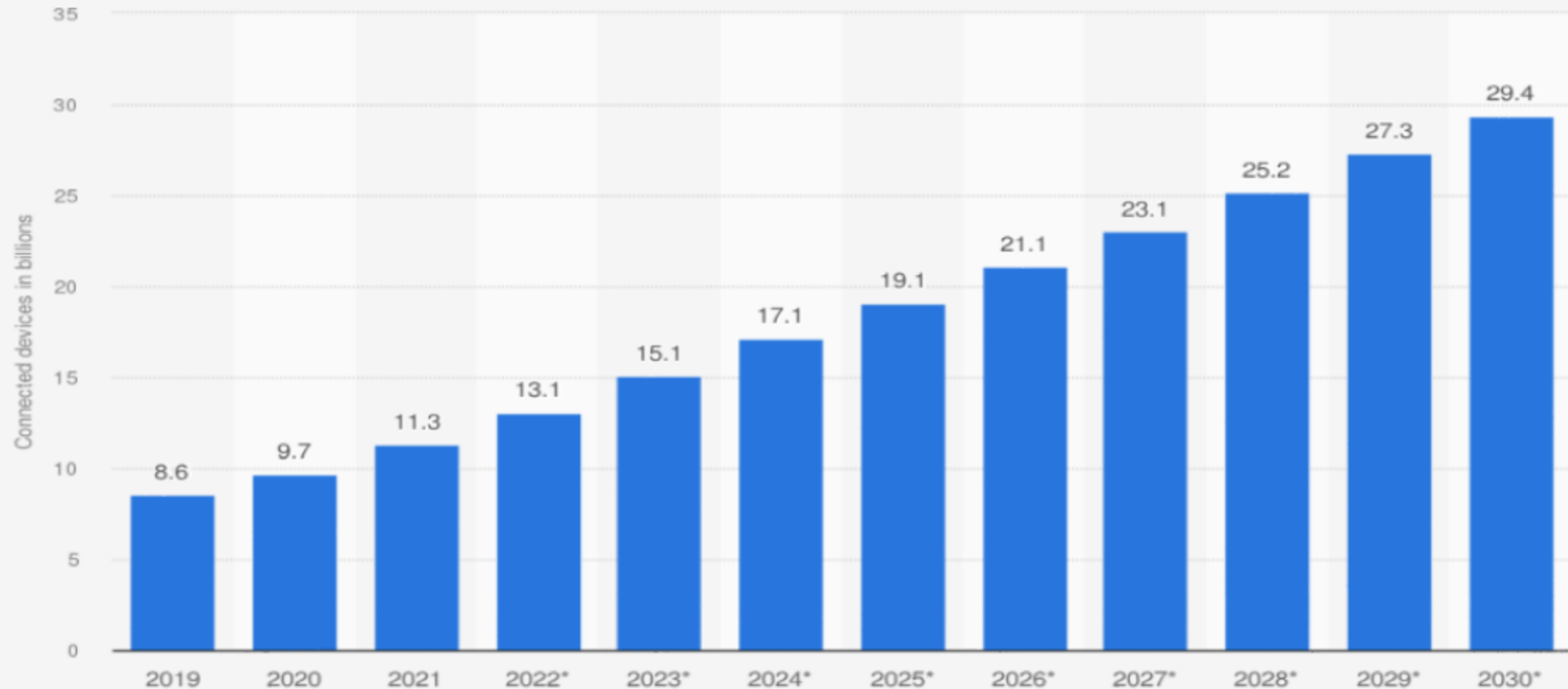


Internet Usage and Population Statistics

5.25 billion people use the internet worldwide. That's **66.2%** of the global population.



Number of Internet of Things (IoT) connected devices worldwide from 2019 to 2021, with forecasts from 2022 to 2030 (in billions)



- The Internet of Things, also called The Internet of Objects, refers to a wireless network between objects, usually the network will be wireless and self-configuring, such as household appliances. - Wikipedia
- The term "Internet of Things" has come to describe a number of technologies and research disciplines that enable the Internet to reach out into the real world of physical objects. – IoT 2008



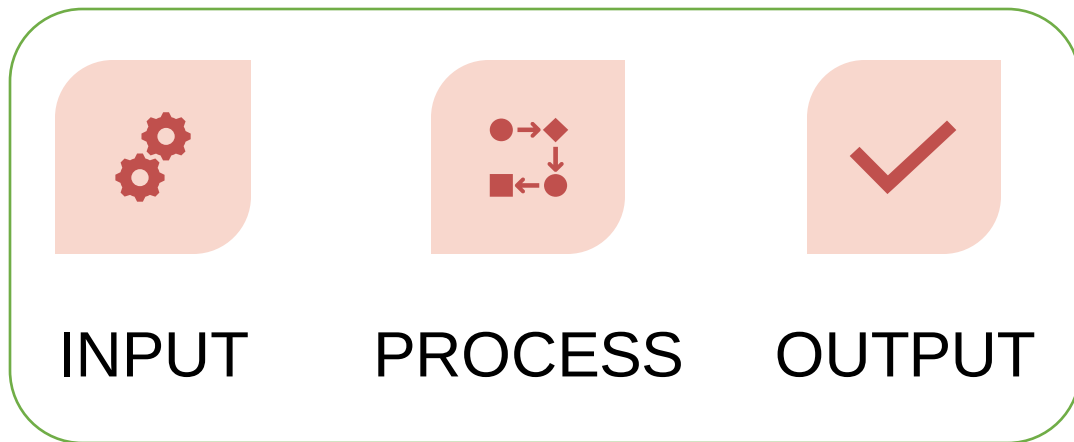
- We want to receive more data
- We want to control stuff
- We want to automate
- We want to make things faster



<http://smartdatacollective.com/rick-delgado/285576/scary-security-concerns-internet-things>

Embedded System – Heart of IoT

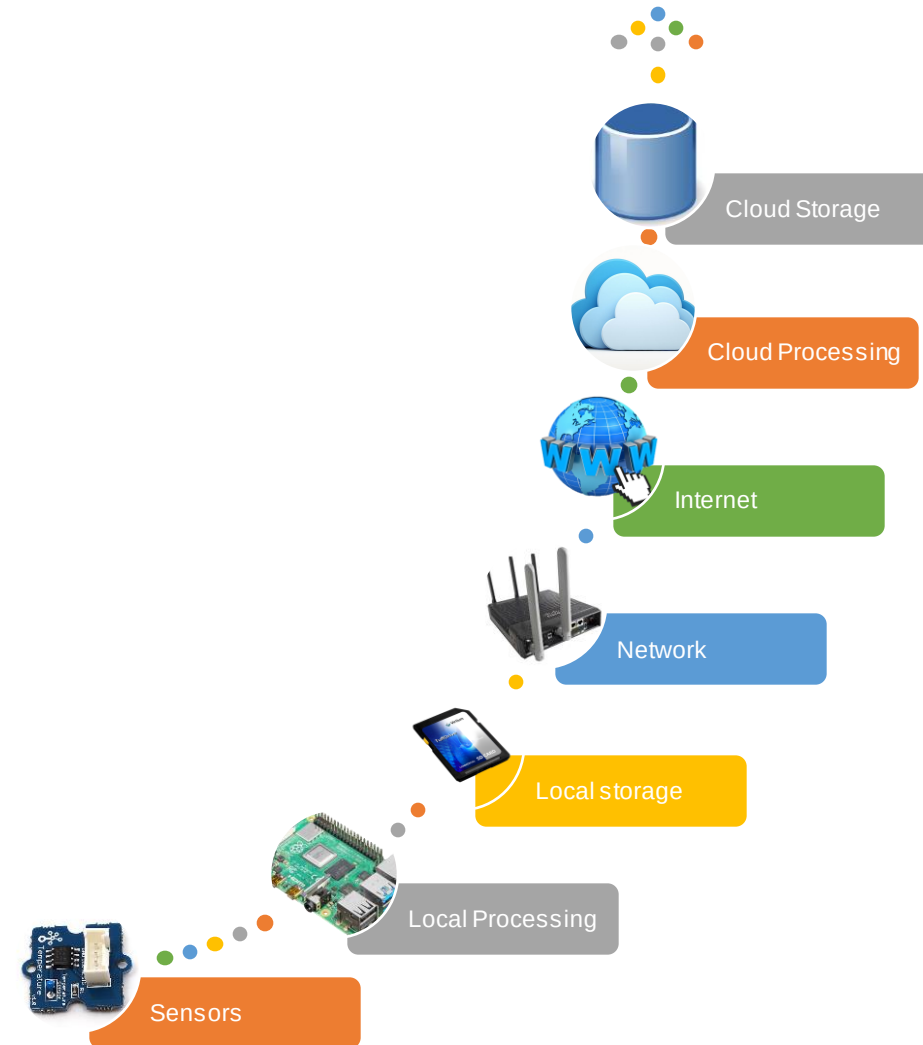
HARDWARE + SOFTWARE



IoT = Embedded System with internet connectivity which enables to share data with other devices used around human environment

IoT Architecture

- Sensors
- Local Processing
- Local Storage
- Network
- Internet
- Cloud Processing
- Cloud Storage



Sensors

- They transform analog data given from scanning the environment to digital data, but they merely do any processing
- Sensor perform following task
 - Measure values
 - Send raw data
 - Low power



Local Processing and Local Storage

- Get data from sensors
- Data Process
- Send some data to Edge/Fog Computing devices



Network and Internet

- IoT Gateway
 - Gathers data from sensors
- Protocols
 - CoAP
 - MQTT
 - HTTP
 - XMPP



Cloud Processing and Storage

- Aggregate Data
- Storage
- Inferences



Top IoT platforms



Cumulocity IoT Platform



Microsoft Azure IoT Suite



Google Cloud's IoT Platform



AWS IoT Platform



Cisco IoT Cloud Connect



Oracle IoT Platform



IBM Watson IoT Platform



**SAP CLOUD PLATFORM
SERVICE: INTERNET OF THINGS**

Where IOT is used? How?



Vehicle, asset, person & pet
monitoring & controlling



Agriculture automation



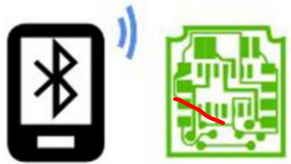
Energy consumption



Security &
surveillance



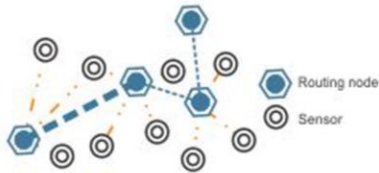
Building management



Embedded
Mobile

Internet of things

Everyday things
get connected  for smarter
tomorrow



M2M & wireless
sensor network



Everyday things



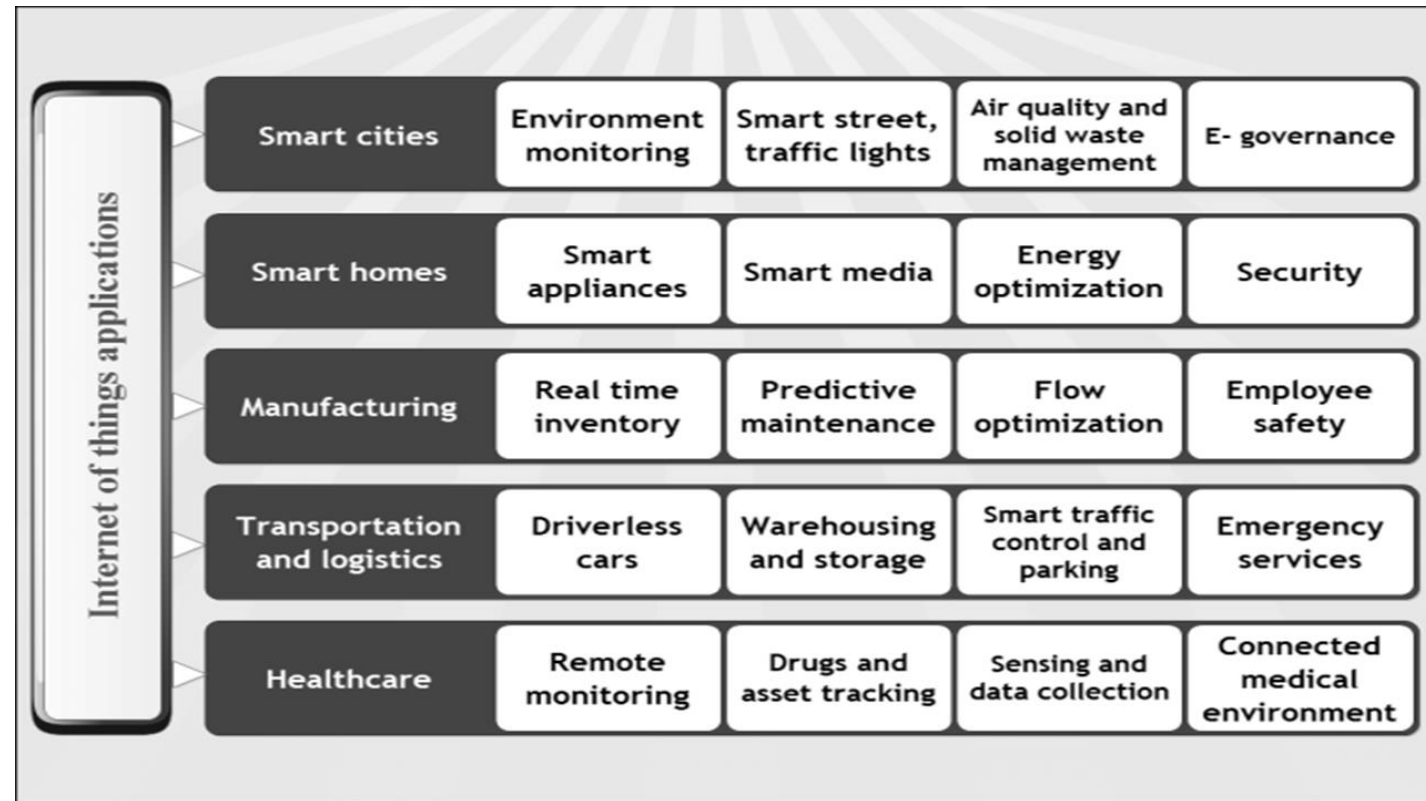
Smart homes & cities



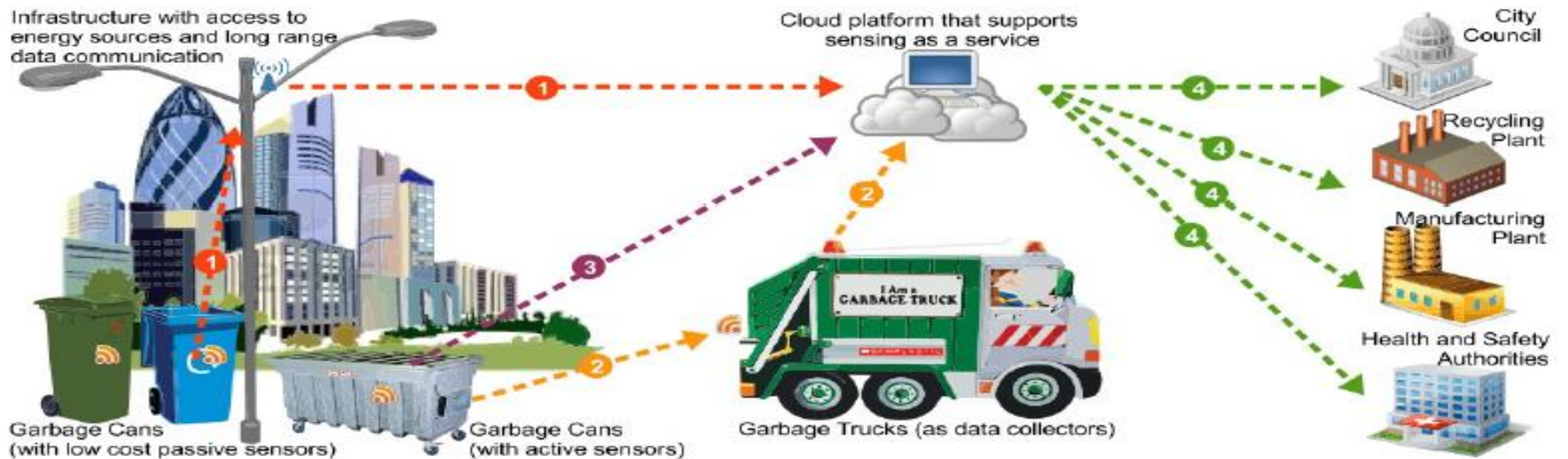
Telemedicine & helthcare

IoT Applications

- Agriculture
- Consumer use
- Health care
- Insurance
- Manufacturing
- Retail
- Transportation
- Utilities/Energy



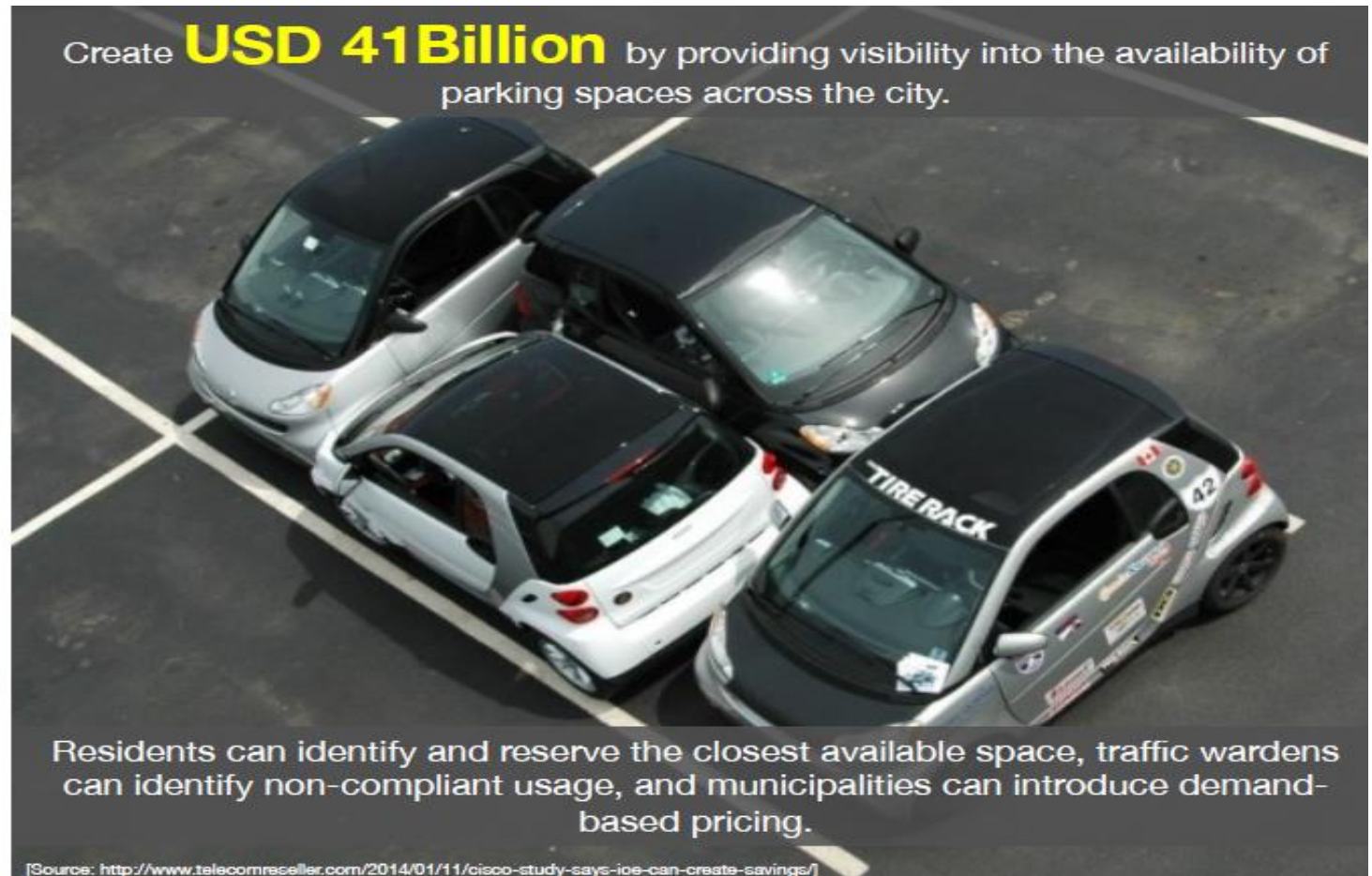
Efficient Waste Management in Smart Cities Supported by the Sensing-as-a -Service



[Source: "Sensing as a Service Model for Smart Cities Supported by Internet of Things", Charith Perera et. al., Transactions on Emerging Telecommunications Technology, 2014]

Smart Parking System

- Create USD 41 Billion by providing visibility into the availability of parking spaces across the city.

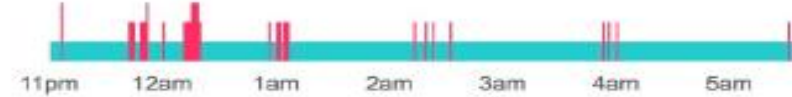


How Well Do I Sleep?

Sleep



Your sleep pattern ■ asleep ■ awake



You went to bed at
11:00PM

Time to fall asleep
0min

Times awakened
20

You were in bed for
6hrs 40min

Actual sleep time
6hrs 6min

8 h 50 mins asleep

- Awake for 212 mins (81x)
- Restless for 278 mins (91x)



Thursday, February 27

Sleep Stats

Time asleep over the past 30 days in hours



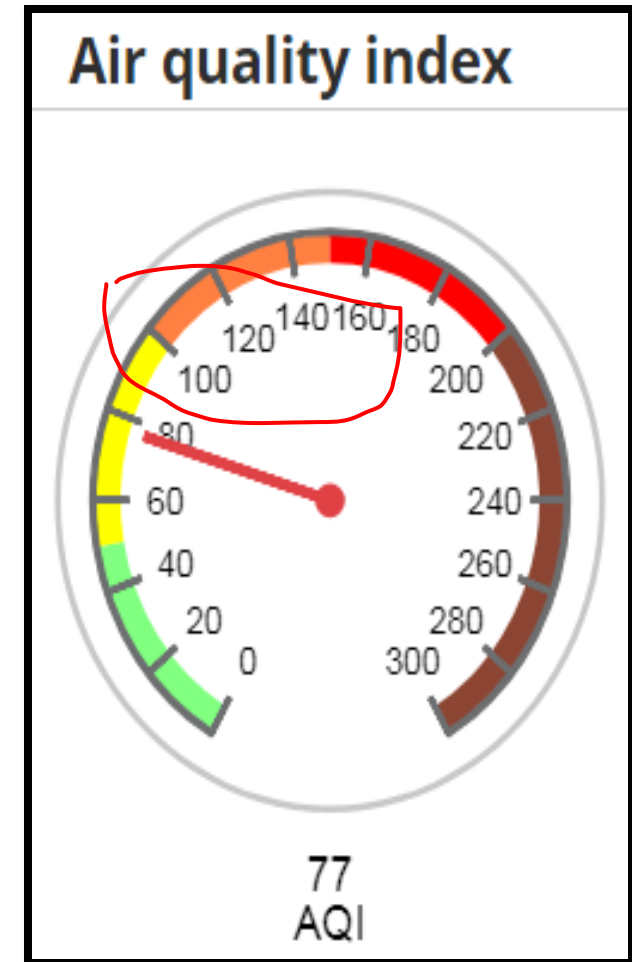
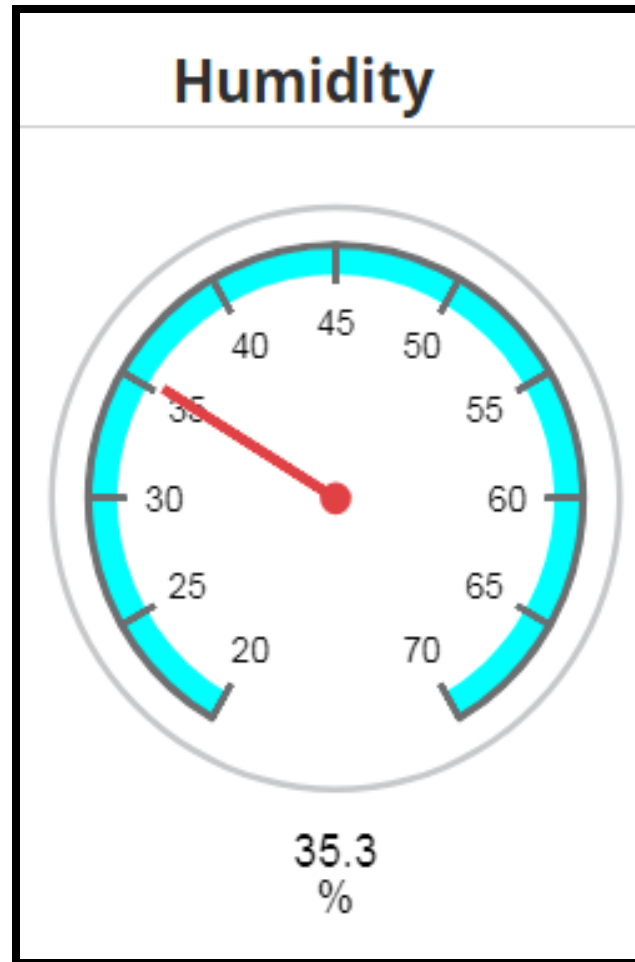
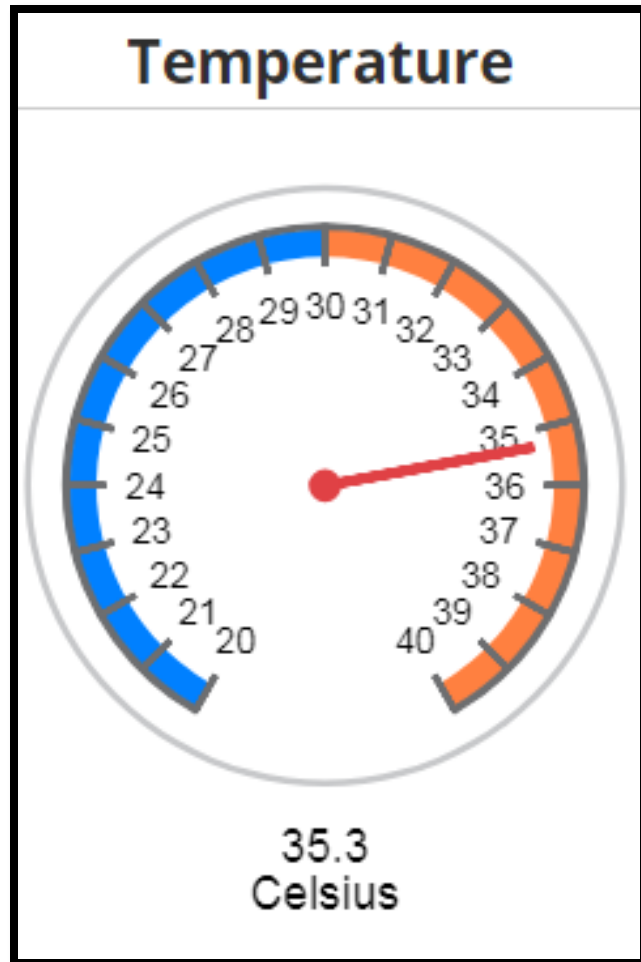
Times awoken over the past 30 days



fitbit flex
Wireless Activity + Sleep Wristband



Weather Monitoring



Limbayat

AQI

0

GOOD

🌡️ 21 °C

💧 64 %

Last updated: in 26443212 minutes

CO2 0 ppm

CO 0 ug/m3

NO2 0 ug/m3

NO 0 ug/m3

SO2 0 ug/m3

PM2.5 0 ug/m3

PM10 0 ug/m3

Varachha

AQI

102

MODERATELY
POLLUTED

🌡️ 35 °C

💧 50 %

Last updated: in 12 minutes

CO2 0 ppm

CO 2.13 ug/m3

NO2 0 ug/m3

NO 37.01 ug/m3

SO2 50.93 ug/m3

PM2.5 48.28 ug/m3

PM10 98.93 ug/m3





Surat Municipal Corporation				
SURAT RAINFALL DETAILS				
Welcome to Surat Municipal Corporation's Mobile Application				
Date/Time		Ukai Dam Outflow (Cuses)	Ukai Dam Level (ft)	Causeway Level (mt)
03/09/2015	20:00	11970.0	333.35	5.32
03/09/2015	14:00	11950.0	333.40	5.32
03/09/2015	08:00	6473.0	333.45	5.32
02/09/2015	20:00	11980.0	333.47	5.33
02/09/2015	14:00	1000.0	333.50	5.34
02/09/2015	08:00	11960.0	333.50	5.34
01/09/2015	20:00	9261.0	333.46	5.34
01/09/2015	14:00	11962.0	333.47	5.34
01/09/2015	08:00	6272.0	333.47	5.34
31/08/2015	20:00	11790.0	333.54	5.34
Ukai Full reservoir Level:345 ft Cusec Way Overflow Level:6.0 mt *Source:Irrigation Dept./Collector Office				





IoT Technology Trends in 2022



Your Global IoT Market Research Partner

10 IoT technology trends to watch in 2022

- 1  IoT is developing into a crucial technology for sustainability
- 2  The platform hype is moving from cloud to the edge
- 3  IIoT initiatives are transforming manufacturing
- 4  Cloud-Native applications are on the rise
- 5  Hyperautomation is transforming operations
- 6  AI is increasingly found at the (Thin) Edge
- 7  "Invisible AI" adoption is happening right under our noses
- 8  Immersive realities (VR/AR) are entering the enterprise environment
- 9  5G is becoming "IoT ready"
- 10  Secure remote access of assets is growing in importance

Source: IoT Analytics Research 2022. We welcome republishing but require source citation with link to the original post and company website.

Reference

- <https://www.broadbandsearch.net/blog/internet-statistics>
- <https://www.cloudcredential.org/blog/knowledge-byte-building-blocks-of-iot-architecture/>
- <https://www.engineersgarage.com/iot-building-blocks-and-architecture-iot-part-2/>
- <https://www.c-sharpcorner.com/UploadFile/f88748/internet-of-things-part-2/>
- <https://www.oracle.com/in/internet-of-things/what-is-iot/>
- <https://www.simplilearn.com/iot-applications-article>